

CSS Syntax - Complete Guide

The Basic CSS Rule

Every CSS instruction follows this pattern:

```
selector {  
  property: value;  
  property: value;  
}
```

Breaking it down:

- **Selector** — Which HTML element(s) to style
- **Property** — What to change (color, size, etc.)
- **Value** — How to change it (red, 20px, etc.)
- **Declaration** — A property-value pair
- **Rule Set** — The selector + all declarations

Selectors

Selectors tell CSS which HTML elements to style.

1. Element Selector

Selects all elements of a type:

```
p {  
  color: blue;  
}  
  
/* All <p> tags will be blue */
```

2. Class Selector

Selects elements with a specific class (use .):

```
.button {  
  background-color: green;  
  padding: 10px;  
}  
  
<button class="button">Click me</button>
```

3. ID Selector

Selects an element with a specific ID (use #):

```
#header {  
  background-color: navy;  
  color: white;  
}  
  
<div id="header">Welcome</div>
```

Note: Each ID should be unique on the page.

4. Attribute Selector

Selects elements based on attributes:

```
/* Select all input fields of type "email" */  
input[type="email"] {  
  border: 2px solid blue;  
}  
  
/* Select all links with href starting with "https" */  
a[href^="https"] {  
  color: green;  
}
```

5. Pseudo-class Selector

Selects elements in a specific state (use :):

```
/* When mouse hovers over link */  
a:hover {  
  color: red;  
}  
  
/* First child of parent */
```

```
li:first-child {
    font-weight: bold;
}

/* When button is clicked (focused) */
button:focus {
    outline: 3px solid blue;
}

/* Visited link */
a:visited {
    color: purple;
}
```

6. Pseudo-element Selector

Selects parts of elements (use ::):

```
/* First letter of paragraph */
p::first-letter {
    font-size: 32px;
    font-weight: bold;
}

/* First line of paragraph */
p::first-line {
    color: blue;
}

/* Before element content */
p::before {
    content: →" ";
}

/* After element content */
p::after {
    content: " ←";
}
```

7. Combinators

Combine multiple selectors:

```
/* Descendant: p inside div */
div p {
    color: blue;
}
```

```

/* Child: p that is direct child of div */
div > p {
    color: green;
}

/* Adjacent sibling: p right after h2 */
h2 + p {
    font-weight: bold;
}

/* All siblings: p after h2 */
h2 ~ p {
    color: gray;
}

```

8. Group Selector

Style multiple selectors together (use ,):

```

h1, h2, h3 {
    color: purple;
    font-family: Arial;
}

/* All headings will be styled the same */

```

Properties and Values

Text Properties

color: blue;	/* Text color */
font-size: 18px;	/* Text size */
font-family: Arial, sans-serif;	/* Font type */
font-weight: bold;	/* Bold (400-700 or bold) */
font-style: italic;	/* Italic style */
text-align: center;	/* left, center, right, justify */
text-decoration: underline;	/* underline, overline, line-through */
line-height: 1.5;	/* Space between lines */
letter-spacing: 2px;	/* Space between letters */
text-transform: uppercase;	/* uppercase, lowercase, capitalize */

Color and Background

```
color: red;                /* Text color */
background-color: yellow;   /* Background color */
background-image: url(...); /* Background image */
background-size: cover;     /* Image size */
background-position: center; /* Image position */
opacity: 0.5;              /* Transparency (0-1) */
```

Box Model: Margin, Padding, Border

```
/* Spacing inside the element */
padding: 10px;           /* All sides */
padding-top: 10px;       /* Top only */
padding: 10px 20px;      /* Vertical, Horizontal */
padding: 10px 20px 30px 40px; /* Top, Right, Bottom, Left */

/* Spacing outside the element */
margin: 10px;            /* All sides */
margin-top: 10px;        /* Top only */
margin: auto;            /* Center horizontally */

/* Border around element */
border: 2px solid black;
border-radius: 10px;     /* Rounded corners */
border-top: 3px dashed blue;
```

Size

```
width: 200px;             /* Fixed width */
width: 50%;               /* Percentage of parent */
height: 150px;            /* Fixed height */
max-width: 300px;         /* Maximum width */
min-height: 100px;        /* Minimum height */
```

Display and Layout

```
display: block;           /* Full width, new line */
display: inline;          /* Only takes needed width */
display: inline-block;    /* Like inline but accepts width/height */
display: flex;            /* Flexible layout */
display: grid;            /* Grid layout */
display: none;            /* Hidden from page */
visibility: hidden;       /* Hidden but takes space */
```

Flexbox

```
display: flex;
flex-direction: row;           /* row, column, row-reverse, column-
    reverse */
justify-content: center;       /* Horizontal alignment */
align-items: center;           /* Vertical alignment */
gap: 10px;                     /* Space between items */
flex-wrap: wrap;               /* Wrap items to next line */
```

Position and Transform

```
position: static;              /* Default */
position: relative;            /* Relative to normal position */
position: absolute;            /* Relative to parent */
position: fixed;               /* Fixed on screen */
top: 10px;                     /* Distance from top */
left: 20px;                    /* Distance from left */

transform: rotate(45deg);       /* Rotate */
transform: scale(1.5);          /* Scale */
transform: translateX(50px);    /* Move */
```

Units in CSS

```
px      /* Pixels (fixed) */
em      /* Relative to font-size of element */
rem     /* Relative to root font-size */
%       /* Percentage of parent */
vh      /* Viewport height (1vh = 1% of screen height) */
vw      /* Viewport width */
pt      /* Points (for print) */
cm, mm  /* Centimeters, millimeters */
```

Colors

```
/* Color name */
color: red;

/* Hex code */
color: #FF0000;
```

```
/* RGB */
color: rgb(255, 0, 0);

/* RGBA (with transparency) */
color: rgba(255, 0, 0, 0.5);

/* HSL */
color: hsl(0, 100%, 50%);
```

Cascading and Specificity

CSS stands for **Cascading** Style Sheets. This means:

```
/* More specific rules override general ones */

/* General rule */
p {
  color: blue;
}

/* More specific - overrides above */
.highlight p {
  color: red;
}

/* Even more specific - overrides both */
#main .highlight p {
  color: green;
}

/* Inline style - highest priority */
<p style="color: yellow;">Text</p>
```

Specificity order (lowest to highest):

1. Element selectors
2. Class selectors
3. ID selectors
4. Inline styles
5. !important (use rarely)

Common Mistakes

```
/* WRONG - Missing colon */
color red;

/* CORRECT */
color: red;

/* WRONG - Missing semicolon */
color: red
background: blue;

/* CORRECT */
color: red;
background: blue;

/* WRONG - Selector syntax */
.class1 .class2 { } /* If not nested */

/* CORRECT - If nested in HTML */
<div class="class1">
  <div class="class2"></div>
</div>
```

Quick Reference

Box Model Order: Margin → Border → Padding → Content

Selectors Priority:

1. Universal selector (*)
2. Element selectors
3. Class/Attribute selectors
4. ID selectors
5. Inline styles

Common Units:

- Pixels (px) for fixed sizes
- rem for responsive design
- % for relative sizing